Tidal potential measured on a coastal–inland optical-clock baseline

Introduction

Earth's gravitational potential changes as the Moon and Sun move, water and air are redistributed, and the solid Earth deforms. Its temporal variations reflect both astronomical forcing and the redistribution of mass in the Earth system.[1] Clocks sense potential through the passage of time: a change in the potential difference between two sites changes their relative ticking rates.[2] A fractional frequency change of 10^{-18} corresponds to about $0.09\,\mathrm{m}^2/\mathrm{s}^2$, equivalent to a height change of roughly 1 cm near Earth's surface. Comparing distant clocks therefore offers a way to observe changes in gravitational potential directly through frequency measurements.

The performance of optical clocks now makes this prospect experimentally accessible. Evaluated fractional systematic uncertainties have reached 8.1×10^{-19} in a strontium lattice clock and 1.2×10^{-19} in a lutetium-ion reference.[3,4] Independent strontium clocks have demonstrated fractional frequency instability of $4.8 \times 10^{-17}/\sqrt{\tau/\mathrm{s}}$, where τ is averaging time.[5] These advances address two distinct requirements: control of systematic shifts determines the accuracy of a potential measurement, whereas stability determines how rapidly a small change can be resolved. Phase-stabilized optical fibres extend comparisons across hundreds of kilometres, and coordinated international campaigns have begun connecting clocks into networks.[6,7]

Clock comparisons have already measured static potential differences, including agreement with conventional levelling and gravimetry, and measurements using transportable clocks.[8,9] Laboratory clock performance has also reached the sensitivity associated with height differences below a centimetre.[10] Together, these developments provide a basis for networks that could monitor time-dependent potential differences and constrain models of mass redistribution.[11] The dynamic measurement poses a further requirement: sufficient precision must be reached while the signal is changing. Its observability depends on the difference between the signals at the two sites, as well as on the stability and availability of the clocks and their connecting link.

Tides provide a physical test of this capability on hourly timescales. The clock-rate signal combines the direct tidal fields of the Moon and Sun with Earth's deformation and the gravitational and displacement effects of ocean loading.[1,12] These contributions have predictable phases but depend on location. They also contain closely spaced rhythms: the principal lunar semidiurnal tide M_2 repeats every 12.42 hours, whereas the principal solar semidiurnal tide S_2 repeats every 12 hours. Their known frequencies differ by about 0.78 μHz. A corresponding lunar maximum arrives about 50 minutes later on successive solar days, so observations on different dates sample different lunar–solar phases. This phase evolution provides a way to distinguish neighbouring tidal responses even when clock operation is interrupted.

Here we measure tidal variations between a strontium clock in coastal Shanghai and a ytterbium clock in inland Wuhan, 692 km apart and connected through Hefei by a 1,350-km phase-stabilized fibre link.[13] The coastal–inland geometry gives the two sites different

responses to ocean mass redistribution. In the combined solid-Earth and FES2022 ocean-loading prediction,[12,14] the M_2 differential fractional-frequency amplitude is 4.78×10^{-18}, corresponding to about $0.43\,\mathrm{m}^2/\mathrm{s}^2$ or an equivalent height of 4.4 cm. Ocean loading contributes about 30% of the combined M_2 phasor amplitude. These quantities set the expected signal scale; equivalent height expresses potential difference, rather than physical displacement of the stations.

The record contains 545.9 hours of observations in 54 segments spread over 353 days. Astronomical phase remains defined at each observation time even when the frequency reference level changes between runs. We retain an independent offset for each segment and compare its internal variations with the predicted tide at the actual observation times. In this way, the interrupted record tests the combined tidal waveform without requiring calibrated frequency-level transfer across interruptions. We then estimate the responses of the M_2+N_2 and S_2+K_2 semidiurnal groups jointly, with relative amplitudes and phases fixed within each group. K_2 contains both lunar and solar forcing. We detect tidal changes in the potential difference and distinguish neighbouring semidiurnal responses within the same clock record.

References

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